How are mass extinctions shaping biodiversity?

Confirmation Review

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July 2025

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Part I

Abstract

Mass extinction events are known to generate profound reorganisations of the global biota. From a macroevolutionary perspective, extinction events are destructive for the lineages that die out, but generative for survivors that are offered ecological opportunity in the form of vacant niches. Adaptive radiation theory suggests that these surviving lineages undergo rapid taxonomic and morphological diversification. The diversification of placental mammals after the Cretaceous-Palaeogene (K-Pg) mass extinction has been traditionally viewed as an archetypal adaptive radiation, with the extinction of the non-avian dinosaurs considered the catalyst for the rapid rise in eutherian diversity (placental mammals and extinct relatives). However, the last decade of discoveries in the fossil record and advances in comparative methods have revealed conflicting evidence surrounding the role of the K-Pg extinction in the taxonomic and morphological diversification of eutherians. Recent studies have suggested an ecological radiation may have begun prior to the K-Pg event, while others indicate significant diversification occurring afterward. These discrepancies in findings are likely rooted in contrasting methodologies when investigating morphological diversity (hereafter termed disparity). The central research question of this project is how different methodological choices, such as the type of morphological data, the phylogenetic tree used and various downstream metrics and models applied, influence our understanding of the effects of the K-Pg event on eutherian disparity. To investigate this, I aim to develop a pipeline that iterates through an extensive number of parameter combinations, reflecting the wide-ranging choices available to researchers studying disparity. Once completed, this pipeline will offer a comprehensive reproducible framework for studying the effects of mass extinctions on disparity, which will elucidate how the discrepancies in adaptive signals during the K-Pg event in previous studies may have been a result of contrasting methodological choices. Furthermore, the pipeline can be a useful tool for the wider community to be applied to other clades and extinction events.

Part II

Literature Review

Lit review needs following structure: - What are big evolutionary questions we want to answer? For example, how do animals evolve across a mass extinction? - Inferring evolutionary events with PCM's/morphology -¿ what are the main methods people use. Vast quantities of data have allowed phylogenies to be easily constructed. - Disparity -¿ what can it be used to infer - However, disparity is not prone to major pitfalls. - Ancestral state estimation -¿ why we use it? lack of fossils etc. Challenges we face when using PCMs.

1 Mass extinctions

1.1 What are they

Extinction is a common process throughout Earth history; 99% of species that have ever existed are now extinct (Overholt, 2003). However, extinction rates are not constant and occur in pulses across background taxonomic turnover (Raup and Sepkoski, 1982). Extinction events that wipe out over 75% of species are termed 'mass extinctions' and have occurred five times over the last 500 million years (Raup and Sepkoski, 1982). Mass extinctions drive a reorganisation of biodiversity as episodes of mass species turnover (Jablonski, 2001). On a macroevolutionary scale, mass extinctions can be considered both destructive and generative; derailing the evolutionary processes of selection and speciation that occur during periods of background extinction (Gould, 2002), yet triggering adaptive radiations in the aftermath as unoccupied niches are filled by surviving clades (Losos, 2010; Toljagić and Butler, 2013). Therefore, mass extinction boundaries often mark the most profound species composition transitions in Earth history.

1.2 Recovery dynamics

Classic adaptive radiation theory posits a linkage between taxonomic and phenotypic diversification, which are expected to follow a pattern of early high rates that decrease through time (Schluter, 2000). However, empirical evidence often finds a decoupling of taxonomic richness and morphological diversity (termed disparity). Peak taxonomic diversity and morphological disparity may be reached at different points in a clade's history, suggesting a discordance between these metrics of diversity (Foote, 1993; Hopkins, 2013; Ruta et al., 2013; Wan et al., 2021). This discordance has been observed across mass extinction events (Bapst et al., 2012; Edie et al., 2018; Ezcurra and Butler, 2018). Therefore, quantifying species composition turnover based solely on taxonomic diversity metrics such as species richness may not encapsulate the full complexity of recovery (Puttick et al., 2020). Survivors filling the vacant niches left by extinct lineages allow this increase in ecological diversity. Alternatively, surviving clades may proliferate without undergoing substantial ecological or functional change by persisting within pre-existing roles (Brusatte et al., 2008; Bapst et al., 2012).

2 Morphology, Disparity and Extinction

Analyses of morphology provide a key insights into past evolutionary processes, especially across major evolutionary events such as a mass extinction. Disparity, the term used to describe morphological variation, has been widely used to investigate patterns of evolution in extant and extinct clades (Wills et al., 1994). Disparity is often used as a way of describing the multidimensional 'trait spaces' that organisms occupy. Guillerme et al. (2020) outlines three ways these traitspaces can be described: size, density and position. Disparity is often used as a proxy for ecological variation, particularly when the morphological traits are inferred to be correlated to ecological function. Mass extinctions pose numerous scenarios of disparity (Figure 1);

intuitively, mass extinctions may be expected to result in a decrease in disparity in tandem with selected removal of taxonomic diversity, as well as suppressed morphological evolution during recovery. Alternatively, when extinction is random, disparity may be maintained. In adaptive scenarios, disparity may increase in opportunistic clades that diversify during the period of ecological opportunity following a mass extinction.

Idea of extinction selectivity: Devonian events are selective, Permian was non-selective (Korn et al., 2013).

2.1 Extinction selectivity

Mass extinction events can have varying effects on the morphospace of clades. By removing a subset of taxa, the distribution of taxa within the morphospace is transformed. Random extinction will reduce the total number of taxa, but have little effect on variance-based disparity metrics, whereas selective extinction, whereby taxa are more susceptible to extinction due to particular trait combinations, will have a detectable effect on variance-based measures (Foote, 1997; Korn et al., 2013). Korn et al. (2013) proposed several patterns to describe different selection processes: lateral (asymmetric selection that trims one side of the morphospace), marginal (symmetric selection that removes peripheral taxa) and non-selective (random process independent of the variables defining the morphospace). In contemporary extinctions, this idea of 'extinction space' is also valid (Ali et al., 2023; Matthews et al., 2024).

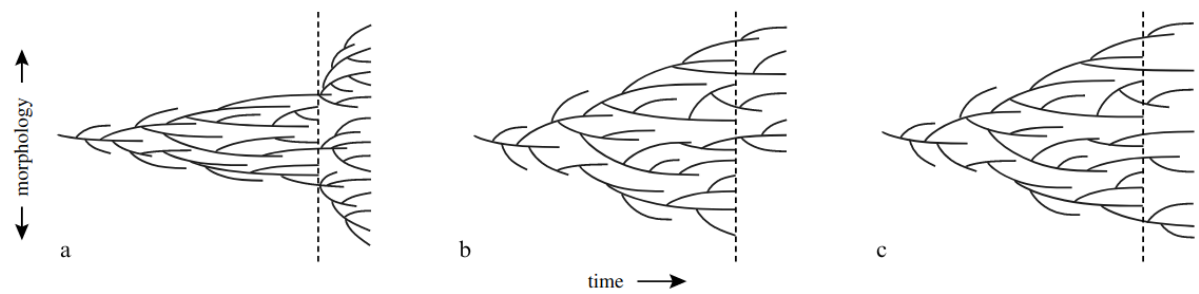

Figure 1: Three scenarios that may occur within a clade across a mass extinction (denoted by dashed line). a) adaptive radiation. b) selective extinction. c) non-selective extinction. Adapted from Grossnickle and Newham (2016).

What do we expect: - disparity to increase across mass extinction according to adaptive radiation theory

2.2 Examples of disparity shifting across mass extinction

Perhaps describe the eutherian mammal issue - no real clue on what really happened here? - mixed evidence of disparity changing across mass extinctions - why is that? - Methodological differences could be the answer?

- One example where animals have radiated after a mass extinction (disparity goes up) - One example where disparity decreases (Cole and Hopkins, 2021).

Can link to Cole and Hopkins (2021).

3 Measuring disparity

3.1 Types of trait data

Morphological data used in disparity analyses usually fit into one of two categories: continuous or discrete. Discrete trait data

3.2 Pipeline

Describe extra step involved in

3.3 Whether or not to include phylogeny

Can link to ancestral state estimation -¿

4 Ancestral state estimation

4.1 Application of ASE

Ancestral state estimation (ASE) is the broad method of estimating trait value(s) in a set of common ancestors given a set of observed data for operational taxa of a phylogenetic tree (Schluter et al., 1997; Cunningham et al., 1998; Yang, 2006; Revell, 2024). Trait data may be continuous, such as body size, or discrete, such as diet type or presence/absence of a character. The application of ASE is rooted in phylogenetic comparative methods, but also has applications in linguistics. As put by Swofford and Maddison (1992), 'character state reconstructions can provide a powerful mechanism for studying many facets of the evolutionary process. However, the zeal with which these techniques are sometimes advocated belies the complexity of the problem'.

Intuitively, ASE is appealing for those studying deep-time patterns. The fragmented nature of the fossil record can yield low sampling coverage at deeper points in a phylogeny and so estimation of node values can increase statistical power. ASE deals with issue of ghost lineages, as brought up by Friedman 2009 and Brusatte 2011. ASE is also needed for doing time-subsampling method of disparity through time.

- Prob

An additional requirement of ASE is the model specified. This model specification varies between discrete and continuous characters.

(Friedman, 2010).

The fragmented fossil record can make sample size at deeper points in the phylogeny low, which may necessitate ASE.

4.2 Discrete characters

Discrete characters are typically analysed using parsimony or likelihood approaches, although likelihood methods are generally preferred. Likelihood approaches, such as Bayesian inference or Maximum likelihood, optimise solutions based on the tree topology, branch lengths and the model of character evolution. Discrete characters are modelled using a continuous time Markov chain process, known as the Mk model (Pagel, 1994; Yang, 2006). This process is specified by a $k \times k$ matrix, $\mathbf{Q}$, which gives the transition rates (q) between states for each character. There are a number of Mk models that can be fit to data. The 'equal-rates model' (ER) is the simplest and estimates just one parameter, since it assumes all transitions occur with equal probability, equivalent to the Jukes-Cantor model of molecular evolution. The assumptions of the ER model can be eased with the 'symmetric' model (SYM): rate of change between any two character states is equal in the forwards and backwards direction, but may differ between different pairs of states. The 'all-rates-different' (ARD) model allows for a different transition rate in each direction, includes a total of $k \times (k-1)$ parameters to estimate.

4.3 Continuous characters

5 Detecting shifts in disparity

6 Aims and objectives

Knowledge gap

Research questions

- How does the accuracy of ASE become filtered down the disparity pipeline?
- How do mass extinction events impact our recovery of ancestral states?
- From the output of the exhaustive permutation of methodological combinations, which of the ecological diversification hypotheses are most supported?

Research objectives

- Build pipeline that iterates over every parameter combination to output hundreds of disparity curves.
- Make this pipeline adaptable so that different types of morphological data can be input, e.g. discrete and continuous characters, landmark data etc.
- Develop pipeline so it uses probability distributions, rather than point estimates, to account for the uncertainty in the input/output.

- Improved methods for ancestral state estimations across mass extinction events.
- Develop new way of measuring change in disparity across a time interval, i.e. quantitative measure of the change in gradient, rather than simply against a null model.

Part III

Thesis plan

7 Chapter One: How is ancestral state estimation accuracy preserved down disparity pipeline?

Empirical wise - can i use a dataset of a well preserved clade eg ammonoids etc.

7.1 Introduction

- Morphology can tell us interesting patterns about evolutionary process blah blah.
- The combination of morphological datasets (using fossils) and phylogenies means we can infer these processes through deep time
- This is particularly useful when studying major evolutionary events (such as a mass extinction)
- However, the fossil record is incomplete and biased.
- Many studies have made use of ancestral state estimation (ASE) in order to increase sample size, infer when certain trait combinations originated and how morphology has changed through time.
- Particularly has been implemented in studies looking at disparity, which describes the multidimensional 'morphospace' in which taxa occupy
- These morphospaces can be constructed using discrete/continuous trait matrices, and ASE can infer the states nodes exhibit using a phylogenetic tree.

7.2 Methods

- I simulated a set of 100 trees using `treats` (Guillerme et al., 2020). These trees were simulated with birth-death paramters as $\lambda = 1.0$, and $\mu = 0.7$. This leaves trees with a range of 50-200 fossil tips.
- I simulated discrete and continuous trait data using `map.traits`.

- For discrete traits, I simulated trait data across each tree (100 characters, 85% binary, 15% 3-state) using Mk model with 3 different transition rates: fast = 0.9, medium = 0.1, slow = 0.01.
- For continuous traits, I simulated both BM and OU model trait data (50 characters).
-
- How does fossil sampling influence ASE accuracy and how does this vary by trait transition-rate/model?
- How does sampling vs threshold improve accuracy of ASE
- How does this accuracy become filtered/distorted down the disparity pipeline?
- How *truly* accurate are certain disparity estimates?
- BONUS how does (selective) extinction influence these estimates?

7.3 Why does this matter?

- ASE is a commonly used method in PCMs
- Used in disparity workflows as a way of increasing sample size in deeper time (where fossil sampling is low)
- Pre-requisite in time-slicing disparity analyses
- Often, studies will use default parameters for ASE functions, without thinking about what to set threshold parameter for example.
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8 Chapter Two: Detecting non-random shifts in disparity across major events (eg extinctions)

9 Chapter Three: Eutherian disparity over K-Pg mass extinction

Part IV

Summary of research achievements

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